Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/01/2017
PI Name:	Wheeler, Raymond Ph.D.		
Project Title:	Dwarf Fruit Production Study		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-2950
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	NASA Exploration Research and Technology		
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PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899-0001	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/20/2015	**End Date:**	01/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17) NOTE: End date changed to 1/31/2017 (original end date 9/30/2016) per E. Connell/M. Whitmore (Ed., 1/21/16)		
Key Personnel Changes/Previous PI:	Co-Investigator, LaShelle Spencer, KSC Engineering Services Contract (ESC).		
COI Name (Institution):	Stutte, Gary Ph.D. (Vencore Inc./NASA Kennedy Space Center) Massa, Gioia (NASA Kennedy Space Center) Spencer, LaShelle E. M.S. (Craig Technologies, Kennedy Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA is planning for long duration missions beyond low Earth orbit. To sustain crew health and performance and prevent micronutritional deficiencies, adequate nutrition must be provided through the food system. The current food system is processed and shelf stable to meet spaceflight requirements, resulting in insufficiencies of several nutrients. One approach to address this is to grow fresh foods in situ, focusing on dietary variety and nutrients that are deficient in packaged food, or that diminish over time with storage. Dwarf tree fruits have never before been studied for use in spaceflight, but fresh fruits are rich sources of vitamins and bioactive compounds and could enhance crew diet by expanding nutritional diversity, and by augmenting flavors, aromas, and overall dietary acceptability. Kennedy Space Center (KSC) solely possesses dwarf cultivars of plums (Prunus domestica) that are expected to be compatible with the volume constrained plant growth hardware on the International Space Station (ISS), such as Veggie. Plums have the potential to reduce loss of bone density (Halloran et al., 2010; Rendina et al., 2013), and to increase the intake of potassium and vitamin K, which are both deficient in the packaged food system. We propose to study the growth and productivity of the dwarf plums in controlled environments under the same environmental conditions found on ISS (light, temperature, relative humidity, and CO2). Specific Aims: 1) Establish dwarf plum trees in controlled environment growth chambers and monitor a range of growth and horticultural parameters using flight constraints (growing volumes, environmental conditions, watering techniques, CO2 levels, and electrical lighting); 2) establish feasibility of growth protocols and cultivar selection for spaceflight, and make recommendations to guide future research solicitations. References Halloran, BP et al. (2010) Dietary dried plum increases bone mass in adult and aged male mice. J Nutr. 140, 1781–1787. Rendina E, Hembree KD, Davis MR, Marlow D, Clarke SL, Halloran BP, Lucas EA, Smith BJ (2013) Dried plum's unique capacity to reverse bone loss and alter bone metabolism in postmenopausal osteoporosis model. PLoS ONE 8: e60569.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This pilot study will gather feasibility data on inclusion of dwarf fruit trees as a food crop for near-term (e.g., ISS) and long duration missions, and on their potential for inclusion in future NRA calls. The study will focus on dwarf genetically modified (GM) plum "trees" that grow from seed or cuttings to fruit in < 1 yr, with no dormancy (cold period) requirements, which is unique to these trees and not typical in other dwarf tree varieties. In field settings, standard sized trees would typically take 5-7 years to begin flowering and fruiting. Kennedy Space Center is the only facility with these dwarf plum trees and it would take multiple years for another institution to even attempt to develop similar plum trees. KSC also has unique on-site plant production facilities, equipment, and controlled environment and spaceflight expertise that enable these trees to be grown under ISS and Veggie growth conditions. This study fits into the AFT PRR Pick and Eat tasks, specifically by supporting nutritional needs. There is a flight NRA planned to begin in FY18 on the AFT PRR, and it is expected that this ground work must be done in the next year to inform an FY17 call on the potential for inclusion of dwarf plum fruits in an ISS study. This would be the first study to assess a dwarf tree for growth in flight rather than a dwarf plant (lettuce, tomato).
Research Impact/Earth Benefits:	Testing with flowering gene (FT1) transformed plum trees has shown these trees no longer have a requirement for cold-period dormancy, which is typical for many members of the genus Prunus (peach, plum, cherry, apricot). If genotypes with desirable traits could be identified, this potentially could expand the range of the FT transformed plums into warmer climates.
Task Progress:	Plum trees (Prunus domestica) that were genetically modified by USDA researchers to overexpress the FT flowering gene were studied as a candidate crop for space food systems. The FT transformation eliminated dormancy requirements and allowed the plants to flower and develop fruit at a very early age (<1 yr), essentially on dwarf plants. Prior horticultural testing at Kennedy Space Center compared the performance of several GM plum genotypes in controlled environment chambers, and four of the better performing lines, NASA-5, -6, -10, and -11, were used for most of the testing reported here. Tests showed that regular irrigations with Peters Excel 13 N – 2 P - 13 K soluble fertilizer produced good growth using commercial potting mixes. But many plants appeared sensitive to having soils too wet or too dry (based on visual assessments). Tests were conducted to improve cloning techniques using branch cuttings dipped in rooting hormone powder (Indole-3-butyric acid —IBA). Studies were also conducted to monitor leaf transpiration and stomatal conductance rates of several genotypes grown at two CO2 concentrations, since CO2 is known to affect stomatal opening and transpiration. The findings showed that stomatal conductance followed a typical rhythm of opening in the light and closing in the dark, but that increasing the CO2 did not reduce conductance, which is typical for most other plants. This may be related to the FT transformation affecting stomatal function and more testing is needed. Elevated CO2 levels showed no adverse effects on growth and potentially could improve growth, but due to the limited number of plants, we were unable to harvest plants for total biomass measurements. Fruit development from these GM plants required cross pollination using wild type pollen to get reliable fruit set. Time from pollination to first sighting of fruit was ~7 days, with time from pollination to mature fruit showing purple color taking about 68 days. Genotypes NASA-5 and NASA-11 were the most reliable for pollination and fruit set. The fruit flesh at maturity was light green, smooth, and juicy. Fruits saved from these and prior tests were analyzed for elemental composition, carotenoids (lutein and zeaxanthin), ORAC antioxidant values, total phenolics, and Vitamin K. Lutein averaged 0.1-0.2 mg / g dry weight, while zeaxanthin was below detection; Vitamin K ranged from 55-75 µg / 100 g dry weight; ORAC from 122 to 257 TE / g dry weight; and total phenolics from 15-23 mg / g dry weight. Plums and prunes are known to slow or reverse bone loss, which could be linked to their high phenolic content and high ORAC values (Smith et al., 2014). Based on some horticultural challenges and need for cross pollination, we recommend that the ptFT plums undergo further testing before consideration for spaceflight. Among the challenges will be testing the potential for launching vegetative cuttings, refining branch pruning techniques to maintain short stature, further breeding to get self-compatible pollination, and further understanding plant water and nutritional requirements, particularly for environments like the ISS, which have CO2 levels of ~2500 ppm or higher. But the effects of the FT genetic transformation of plums are dramatic and should be explored with other fruit trees to bring them into the discussions for future space food crops.

	Reference Smith, B. J. Bu SY, Wang Y, Rendina E, Lim YF, Marlow D, Clarke SL, Cullen DM, Lucas EA. A comparative study of the bone metabolic response to dried plum supplementation and PTH treatment in adult, osteopenic ovariectomized rat. Bone 58, 151–159 (2014).
Bibliography Type:	Description: (Last Updated: 02/08/2019)
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